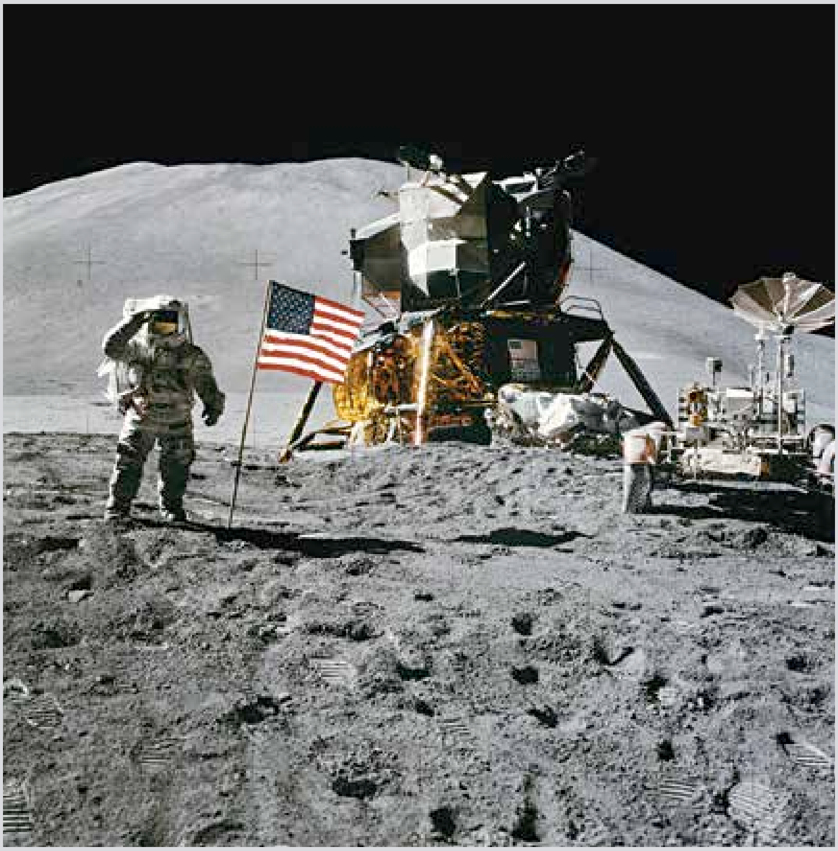
believe that we should not simply waste our time and energy (to say nothing of the resources) on such explorations while we are yet to solve the big social problems on Earth -- the upcoming energy crisis, economic instability, general unrest and the ilk. Both are correct to varying degrees, former probably a little more than the latter, but both have valid points. But a very select group of people get that there is another way to look at the situation. Space is called the 'final frontier' with good reason. Innovations to conquer the constraints placed on us in space travel have fostered an astonishing number of innovations that we take for granted today. From artificial limbs to fire-fighting equipment, the range of sciences that have been touched by this pursuit is impressive.

They say one of the greatest virtues of mankind, one that has kept us physically rather unremarkable beings at the top of the planets metaphorical food chain for so long, is that we are an inherently ambitious species. We've moved far beyond being satisfied by the basic urges and necessities of life. Having conquered Earth, for all intents and purposes (unless Dolphins really turn out to be as smart as Douglas Adams said they were), it is only logical that we now look to the heavens. For years, when we had no ways of reaching them, humanity looked up to the skies and assumed that the Gods lived there, because we couldn't reach those realms. But lo and behold, here we are, ready to board the flight to the red planet.

The Kardashev Scale divides civilizations into three distinct types. We, the ones harnessing the energy of our home planet and nothing else, are the lowly Type I plebs. Type II can harness its star(s), while the final type can harness its own galaxy. Given enough time, a civilization that exists should evolve to become Type III, and yet we see none around. That is to say, there must be 'a great filter' to the jumps in type on this scale, the one that civilizations cannot cross, and are doomed to stagnate and eventually die. We're not sure if we've already crossed the filter, or if it still remains as a challenge for the future. But one thing is for certain -- the pace at which we are currently moving, we most definitely are not hitting stagnation any time soon. d

WHAT WE GAINED BY THE APOLLO PROGRAM



To make a long story short, the Apollo series of missions by NASA were fundamental to the incredible technological leap made by humans over the course of these past 50 years. Started (allegedly) as another pawn in the Cold War race for supremacy, the program laid the seeds for much of our modern accomplishments.

By the time their activities died down, the Apollo missions had brought almost 400 KGs of moon rock onto the Earth. A large part is still stored across various laboratories across the world, and continues to be a great aid in geological, astronomical, and biological research. One of the greatest accomplishments that came from this is that scientists could finally agree on a single theory for the existence of the Moon (separated from Earth by an asteroid impact). That in turn put a lot of theories in their place and allowed us to form a good model of our understanding of the history of the Solar System, and also of the Universe. Of course, a lot of the major advances in the world of science was a by-product of the innovation poured on this set of projects. Fireproof materials, now taken for granted in most modes of long-distance transport, chemical control in water and atmosphere, water purification, pollution remediation, fuel cells, the list can go on. But the gist of the matter is that the these amenities, staples of modern life due to reduction in cost by years of research, were first pioneered by programs such as this one.

All of that, and we haven't taken into consideration what it achieved in spirit. Inspiring a generation of young, aspiring minds, it convinced them that there really was no limit to what we can do. Countless scientists, researchers, and engineers attribute their initial interest into the field by the hype that went through the world when man set forth and stepped on the moon. One giant leap for mankind, indeed.